

THE STRUCTURE OF MATHEMATICAL REALITY

Phase III — Chapter 8

Vector Spaces and the Meaning of “Canvas”

Assignment

Self-Study Curriculum

Take a concrete data analysis problem you have worked on. Write a one-page structural analysis that covers:

1. What vector space your data lived in (be specific about the dimension and what the components represent)
2. What basis you implicitly chose (and what alternatives existed)
3. What inner product you implicitly used (and how it defined your notion of “similarity” or “distance”)
4. Whether the analysis was linear or nonlinear, and if nonlinear, whether a different canvas would have made it linear
5. What the eigenvalues/singular values told you, stated in terms of the system’s structure rather than the mathematical computation